MOONFIRE VENTURES HACK · AUTORESEARCH FOR REAL-WORLD VALUE

When to pull an engine

An autoresearch loop that tunes engine removal against the bill an airline actually pays.

Early is wasted engine life. Late is a grounded aircraft.

[Team name] · [Names]

Every engine pull is a bet on time

PULL TOO EARLY

Wasted life

Every cycle left on the wing was paid for. Pull too soon and it goes in the bin, engine after engine, across the fleet.

PULL TOO LATE

$50k to $150k

One estimate for a single day on the ground after an engine event. Three days or more can pass $600,000.

Boeing puts a grounded aircraft as high as $150,000 an hour. The right day to pull sits between the two mistakes.

Sources: Air Cargo Week, “The real cost of AOG”; Boeing estimate cited in Wikipedia, “Aircraft on ground”.

THE CLAIM

Pulling an engine is a threshold on a forecast. Both can be searched, and both can be scored against the airline’s own bill.

That is verifiable autoresearch with money on the line.

One turn of the loop

01

Hypothesis

Read the log and write one line on what should help.

02

Edit

Change model.py, the only file the agent can touch.

03

Run

Train on four folds of engines, replay the fifth, five times over.

04

Score

Mean cost per engine, failures first, waste second.

05

Keep or revert

Commit if cost falls and it wins four folds of five. If not, reset. Repeat.

Every turn logged to results.tsv · about a minute a turn · about 30 turns an hour on a laptop CPU

What the agent can and cannot touch

Open: model.py

Features, like rolling means and slopes over recent cycles

Which of the 21 sensors to keep

A cap on target life, near 125 cycles

The model, and a loss that charges late errors more

The pull threshold T

Locked

The split, by engine and never by row

The scorer and the cost of a failure

A 60-second limit on every run

The test engines and the second fault mode, sealed until the end

The score is the airline’s bill

Replay walks each held-out engine forward one cycle at a time, as an airline would live it, and pulls the engine the first time predicted life falls to T or below.

A failure is priced above any engine’s whole life, so the agent can never win by letting engines fail. The score reads failures first, waste second.

We also report NASA’s asymmetric score, which punishes late warnings harder than early ones, and RMSE, so results sit beside published work.

PULLED EARLY

`cost = cycles of life left`

RAN TO FAILURE

`cost = 500 cycles`

SCORE TO MINIMISE

`mean cost per engine`

500 is our assumption. An airline sets its own ratio, and we rerun at 250 and 1,000 to show the ranking holds.

Data: NASA C-MAPSS turbofan degradation set, subsets FD001 and FD003.

Four ways the loop cannot cheat

Risk	Guard
Memorising an engine	Split by engine, never by row, so no engine sits in both training and scoring
Peeking at the future	Each track is cut at a random cycle, and predictions before the cut must not change
A lucky fold	A change must win on four folds of five, not on the average alone
Learning the benchmark	Test engines and a second fault mode stay sealed until one final run

What the hour produced

COST PER ENGINE

[___]

Baseline [__], lower is better

RUN TO FAILURE

[___]

Of 100 replayed engines, baseline [__]

CHANGES KEPT

[___]

Out of [__] turns, each one a commit

Sealed check, run once	Baseline	Best
FD001 test, NASA score	[__]	[__]
FD001 test, RMSE in cycles	[__]	[__]
FD003, unseen fault mode, NASA score	[__]	[__]
Ranking holds at 250 and 1,000 cycles	n/a	[yes or no]

How it plugs into an airline

DATA IN

The airline’s own engine health data takes the place of C-MAPSS: sensor snapshots per engine per flight, with removal records as the labels.

WHAT COMES OUT

A model, a pull threshold and a log of every change the agent tried, why it tried it and what it scored.

HOW IT RUNS

On a laptop CPU, rerun each week as new removals arrive. A reliability engineer signs off each kept change.

The cost of a failure against a cycle of life is a business input the airline sets, not a number hidden in our code.

WHEN TO PULL AN ENGINE

The loop finds the day between wasted life and a grounded aircraft.

Built on public data in an hour. Ready to point at an airline’s own.

[Our ask]

[Team name] · [Contact]